Exploring uncertainty in Antarctica's future sea level contribution

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Total word count: 7994 words.

Chapter 1

Literature Review

1.1 Sea level change

540 words

The IPCC's Sixth Assessment Report (AR6) identified sea level rise as "an existential threat for island nations, low-lying coastal zones, and the communities, infrastructure, and cultural heritage therein" (Cooley et al., 2022). A rise of 1 m would increase the global population living below mean sea level from between 34–49 million people today to 77–132 million (Seeger and Minderhoud, 2026). An elevated mean state also increases the frequency and magnitude of extreme sea level events — episodic departures from the mean driven by tides, storm surges, and tsunamis (Woodworth et al., 2019; Boumis et al., 2023). Exposure to sea level rise is highly regionalised, shaped by vertical land movement and population density (Nicholls et al., 2021; Rentschler et al., 2023). These communities, concentrated in the Global South (Seeger and Minderhoud, 2026), can also be the least economically and institutionally equipped to adapt (O'Neill et al., 2022).

Since the 20th century, global mean sea level rise has accelerated from 1.35 ± 0.57 mm yr^{-1} between 1901–1990 to 3.69 ± 0.48 mm yr^{-1} between 2006–2018 (Fox-Kemper et al., 2021). AR6 gave *high confidence* that this acceleration is the result of anthropogenic warming and that future sea level rise is sensitive to future greenhouse gas emissions. AR6 projects a likely (66% chance) sea level rise of 0.33–0.62 m by 2100 under a low emissions pathway and 0.63–1.01 m under a high emissions pathway, the upper bound of which coincides with the 1 m sea level rise hypothetical evaluated by Seeger and Minderhoud (2026). Sea level rise occurs either through thermosteric effects ("thermal expansion") or through barystatic contributions (i.e. mass flux) from another source such as land water storage, mountain glaciers, or the polar ice sheets. The dominant drivers of historic sea level rise have been thermal expansion and mountain glaciers. However, contributions from the ice sheets have accelerated since the 1970s and are projected to be significant sources of sea level rise this century (Fox-Kemper et al., 2021). Antarctica is the largest source of uncertainty in these projections, due in large part to poorly constrained processes governing the ice sheet response (Edwards et al., 2021). **NOTE: I'm uncertain about the etiquette for citing the same report multiple times in one paragraph. Have I done this appropriately?**

Beyond 2100, AR6 cites "deep uncertainty" in multi-century sea level projections. The 21st century projections are informed heavily by model intercomparison projects, but no such project existed for multi-century projections at the time. Instead, AR6 relied on a combination of methods, including extrapolation of 2100 rates of mass loss and the expert elicitation by Bamber et al. (2019). They also considered Marine Ice Cliff Instability (MICI), a highly uncertain process related to the failure of tall ice cliffs, which was only included in a handful of modelling studies at the time, but which dramatically increase Antarctica's mass loss on these timescales (DeConto and Pollard, 2016; DeConto et al., 2021). Even excluding this process, AR6 projected 0.3–3.1 m of sea level rise by 2300 under a low emissions pathway and 1.7–6.8 m under high emissions. Synthesized estimates for the individual sources were not provided, but uncertainty in Antarctica's contribution was wider than 3 m in some assessments of the high emission pathway. Understanding and constraining this uncertainty is therefore a significant goal towards more confident multi-century sea level projections. Since AR6, an increased number of studies have simulated the Antarctic response to multi-century climate change (see Section 1.5.3), but constraining the uncertainty in its sea level contribution remains a challenge, thus demonstrating the need for further work.

1.2 The Antarctic Ice Sheet

1.2.1 Geographical setting

592 words

The Antarctic Ice Sheet is the largest store of freshwater on Earth, holding 26.0 ± 0.4 million km^3 of ice (Morlighem et al., 2020). It is a dynamic ice mass, accumulating ice through precipitation and flowing outwards under gravity. This flow organises into ice streams, which are fast-flowing corridors of ice that transport ice from the ice sheet interior to its perimeter (Bennett, 2003). 93% of the ice sheet perimeter interfaces with the ocean, either as a floating ice shelf (74%) or vertical ice cliff (19%) (Bindschadler et al., 2011). Ice cliffs discharge ice mainly through the calving of icebergs (Benn and Åström, 2018) and ice shelves through a combination of calving and basal melting (Davison et al., 2023). The transition point between grounded ice sheet and floating ice shelf is called the "grounding line", although in practice this is challenging to define and measure (Friedl et al., 2020). Ice shelves don't contribute directly to sea level rise, but have indirect consequences for sea level through the buttressing force they impose on upstream grounded ice (see Section 1.3.4).

Antarctica can be split into three climatically distinct regions: West Antarctica, East Antarctica, and the Antarctic Peninsula. The West Antarctic Ice Sheet (WAIS) is grounded on bedrock that lies predominantly below sea level (a marine ice sheet), which makes it susceptible to instabilities that could bring about self-sustaining mass loss (see Section 1.3.6). This is particularly concerning because West Antarctica is the region with the highest current rate of mass loss (Table 1.1) and may have collapsed completely in past warm climates (see Section 1.2.3). West Antarctic ice streams drain into the Amundsen Sea, Ross Sea, and Weddell Sea sectors of the Southern Ocean. The latter two feature Antarctica's two largest ice shelves, the Ross ice shelf (**TODO: value** km^2) and the Filchner-Ronne ice shelf (**TODO: value** km^2), respectively. The ocean cavities beneath them are key regions for the production of dense deep waters that form part of the global ocean circulation (Rintoul et al., 2025).

Table 1.1: Antarctica's component ice sheets, their rate of mass loss between 1992–2020 (Otosaka et al., 2023), and their total sea level potential (Morlighem et al., 2020).

Region	Mass loss (Gt yr^{-1})	Sea level potential (m)
WAIS	-82 ± 9	5.3 ± 0.2
EAIS	3 ± 15	52.2 ± 0.7
APIS	-13 ± 5	0.27 ± 0.02

The East Antarctic Ice Sheet (EAIS) is separated from West Antarctica by the Trans-Antarctic mountains. It represents by far the largest share of Antarctica's sea level potential, but has contributed the least to recent sea level change (Table 1.1). Despite the current trends, the EAIS may have been more dynamic in past climates and its future response is highly uncertain (Stokes et al. 2022). Although the EAIS is mostly terrestrial (grounded above sea level), its underlying bedrock has three key marine basins: The Recovery, Wilkes, and Aurora subglacial basins. The Totten glacier, which drains most of the Aurora subglacial basin, alone holds 3.89 m sea level equivalent of ice (Morlighem et al., 2020). These basins may be susceptible to the same instabilities as West Antarctica if they are sufficiently pushed from their current state (Mengel and Levermann, 2014; Aitken et al., 2016; Golledge et al., 2017a).

The Antarctic Peninsula Ice Sheet (APIS) is a comparatively small part of Antarctica, but is losing mass faster than any other region relative to its size (Table 1.1). It is distinct as the most Northerly region of the continent, reaching 63°S towards the Drake Passage. Its mountain range features smaller, Alpine-like glaciers down steep terrain. The Peninsula acts as an orographic barrier between the Weddell Sea and the Bellinghausen Sea, strongly influencing the climate on either side (Davies et al., 2026). The largest ice shelf in the APIS, the Larsen ice shelf, extends down the Eastern flank of the peninsula and is divided into smaller sections Larsen A to D, the largest of which is Larsen C at **TODO: value and cite** km^2. The recent changes in the Antarctic Peninsula are significant not only to the local environment, but also serve as a bellwether for processes that may later occur in East and West Antarctica with substantially greater consequences for sea level.

1.2.2 Recent observed changes

Antarctica is not in equilibrium with the current climate, with some areas showing a clear trend in mass balance that is consistent across multiple methods of measurement (e.g. Gardner et al. 2018; Smith et al. 2020; Velicogna et al. 2020). Furthermore, mass loss from West Antarctica has accelerated across every four year period from -37 ± 19 Gt yr^{-1} in 1992–1996 to -131 ± 21 Gt yr^{-1} in 2012–2016, before slowing to -94 ± 25 Gt yr^{-1} in 2017–2020 (Otosaka et al., 2023). The vast majority of this mass loss has originated from the Amundsen Sea Sector of West Antarctica, particularly the Thwaites and Pine Island Glaciers (Rignot et al., 2019). Sediment records from the Amundsen Sea indicate that Thwaites and Pine Island have been in disequilibrium since at least the 1940s **TODO: cite**. Direct measurements of Pine Island Glacier made possible by the satellite era reveal a consistent pattern of change: retreat of the glacier terminus, acceleration of its ice stream, and a subsequent thinning as the glacier adjusts to maintain its flux balance (Joughin et al., 2003; Payne et al., 2004) **TODO: cite more recent**. It is difficult to draw direct links between the current disequilibrium and anthropogenic climate change, but it's likely that climate change has at least exacerbated the retreat of Pine Island glacier over this period (Bradley et al., 2025). **TODO: See Ines's 2023 remote sensing review paper for more inspiration on Pine Island**.

TODO: Thwaites

Unlike the Arctic, which has warmed by **TODO: value** times the global average due to nonlinear feedbacks (Previdi et al., 2021), Antarctic surface air temperatures have broadly followed the global trend **TODO: cite**. The one exception to this is the Antarctic Peninsula, which has warmed three times faster than the rest of the continent, up to temperatures otherwise unprecedented in the Holocene (Vaughan et al., 2003). This atmospheric warming has increased surface melting of the Antarctic Peninsula Ice Sheet and its resulting sea level contribution (Vaughan, 2006). The most pronounced changes in the APIS have been the thinning, retreat, and collapse of some of the peninsula's ice shelves (Cook and Vaughan, 2010; Rignot et al., 2013). The extent of ice shelves on the peninsula roughly trace the -9°C isotherm, suggesting a "limit of viability" for APIS ice shelves that will shrink with further warming (Morris and Vaughan, 2013). Although this is a useful heuristic, the mechanisms of ice shelf collapse are not purely thermal (see, for example, Sections 1.3.3 and 1.3.4).

The first ice shelf collapse to occur during the satellite era was the Wordie Ice Shelf in 1989 (Doake and Vaughan, 1991). This was followed by a sequential collapse from North to South down the Eastern flank of the peninsula, starting with the Larsen inlet — also in 1989 — then the Prince Gustav and Larsen A ice shelves in 1995 (Cook and Vaughan, 2010). Finally, the Larsen B ice shelf progressively thinned between 1992 and 2001 before its eventual collapse over a few days in February 2002 (Shepherd et al., 2003). The Larsen C ice shelf has so far avoided collapse, but has continued to thin at a rate of -3.8 m per decade over increasingly large areas of the ice shelf (Paolo et al., 2015). Between August 2015 and July 2017, a 200 km crack developed along the ice shelf, leading to the calving of a large iceberg representing 10% of the total ice shelf area (Hogg and Gudmundsson, 2017). Due to the loss of buttressing force provided by ice shelves (see Section 1.3.3), the collapse of each ice shelf on the peninsula was consistently followed by an acceleration and thinning of its upstream glaciers (Rignot et al., 2004; Glasser et al., 2011; Dømgaard et al., 2025). The collapse of the Prince Gustav and Larsen A ice shelves cannot be uniquely attributed to anthropogenic climate change, since sediment records indicate that they have previously collapsed during the mid-Holocene (Pudsey and Evans, 2001; Brachfeld et al., 2003). This is corroborated by written accounts of the Prince Gustav ice shelf by 19th century polar explorers suggesting that it had been in retreat since at least 1843 (Cooper, 1997). However, similar records from ocean sediments previously beneath the Larsen B ice shelf indicate that it had been stable throughout the Holocene, and therefore suggest that its recent collapse contains an anthropogenic signal

(Domack et al., 2005).

NOTE: Too much on the Peninsula? Two big paragraphs on the smallest region (and least important for my work) seems like a lot. On the other hand, the peninsula is an excellent case study for what might happen elsewhere in Antarctica, an argument which I plan to develop in later sections, so it is perhaps worth going into detail here.

In contrast to the WAIS and APIS, the East Antarctic Ice Sheet remains close to a state of balance, with a slight positive mass trend of 3 ± 15 Gt yr^{-1} between 1992–2020 (Otosaka et al., 2023). The high uncertainty in this estimate is the result of disagreement between different methods of measurement, with some studies showing a negative mass trend over similar periods (e.g. Rignot et al. 2019; Shepherd et al. 2019). These aggregate measurements also belie the spatial variability in East Antarctic mass trends. Several studies highlight regional mass loss from Wilkes Land in East Antarctica, particularly the Totten Glacier, which drains the Aurora subglacial basin (Shepherd et al., 2019; Velicogna et al., 2020; Smith et al., 2020). This is paired with a positive mass trend over the East Antarctic plateau, consistent with reanalysis-derived increases in precipitation (Davis et al., 2005). This spatial variability suggest that whilst the East Antarctic Ice Sheet is in a state of balance, it is not in a state of equilibrium, with the potential for more significant mass change if this balance is disrupted.

TODO: Check Jamie's and Mira's theses to look for more EAIS notes.

TODO: Quick synthesis of this section?

1.2.3 Antarctica in past climates

972 words

NOTE: Comparing this to Jamie's section on Antarctica in the Pliocene,

mine is much more condensed and missing quite a lot of context. However, I wouldn't expect my section to be as big as Jamie's anyway, given the relative importance of the Pliocene in my thesis. For example, I essentially pick up with Pliocene sea level estimates in 2015, whereas Jamie gives a longer history. I should probably include some discussion of palaeoshoreline work, which gave even higher sea level estimates than d18O. I could also spend time explaining how the δ^{18}O record works. I generally feel that I am rushing through references without giving them proper explanation or room to breath, but this may also be a matter of taste.

Past climates are a natural laboratory for exploring the how the Earth system responds to changes in greenhouse gases. By comparison with the modern era, they can inform projections of the future climate (Tierney et al., 2020). The Mid-Pliocene Warm Period (mPWP), a period 3.3–3.0 million years ago with CO_2 concentrations near 400 ppm and temperatures 3–5.3°C warmer than present (Tierney et al., 2025), is the closest analogue to end of the century projections under medium to high emissions (Burke et al., 2018). **NOTE: I've used the Tierney 2025 numbers here because they seem quite convincing, but perhaps I should stick to the IPCC values and simply mention that it could be warmer?** Early attempts to estimate Pliocene sea level based on the the δ^{18}O record — preserved in the layers of benthic foraminifera that occupy ocean sediment cores — produced highly uncertain results (Dutton et al., 2015). For example, Miller et al. (2012) estimated that sea level was 22 ± 10 m higher than today, the lower bound of which would require the near total loss of the Greenland and West Antarctic Ice Sheets **NOTE: This is just technically false because miller et al use a multiproxy approach, not just d18O**. The upper range of these estimates then requires significant contribution from marine and possibly terrestrial sectors of East Antarctica. Much lower sea level estimates can be obtained by using a higher δ^{18}O of Antarctic ice, justifiable by warmer temperatures of the Pliocene (Winnick and Caves, 2015; Gasson et al., 2016), but the true value is then highly uncertain. Raymo et al. (2018) highlight further biases in the δ^{18}O record that could both under- and over-estimate Pliocene sea level.

NOTE: I've again condensed a lot of information below compared to Jamie's. The references I've cited could all be explained in much more detail, but perhaps this level of detail is appropriate for my thesis?

Regional palaeoclimate records provide a suite of evidence for a dynamic WAIS that advanced and retreated in obliquity-paced glacial-interglacial cycles: Naish et al. (2009) and Seidenstein et al. (2024) found open-ocean diatom species and warm-water foraminifera in sediments beneath the current Ross ice shelf; Gohl et al. (2013, 2021) reveal buried grounding zone wedges in various locations of the stratigraphic record; and Passchier et al. (2025) and Horikawa et al. (2026) demonstrate a cyclicity in the sediment composition of the continental shelf. In contrast, evidence for change in the EAIS is more fragmentary, but an increasing number of palaeoclimate records point to periods of substantial retreat. Sediment cores offshore of the George V and Adélie coasts show a sequence of change in ocean productivity, ice-rafted debris, and sediment provenance pointing to retreat in the Wilkes subglacial basin (Cook et al., 2013; Patterson et al., 2014; Bertram et al., 2018). Ice rafted debris from Wilkes land and Adélie land has been found as far away as Prydz Bay, which would require large icebergs to transport, perhaps comparable to Northern Hemisphere Heinrich events (Williams et al., 2010). This is supported by deep glacial erosion representing a plausible location for the terminus of Totten Glacier in the Pliocene, 150 km behind its current grounding line (Aitken et al., 2016). Conversely, seismic soundings of the Aurora subglacial basin evidence the existence of ice near current grounding lines since the late Miocene (Gulick et al., 2017). The EAIS also appears to have responded differently to orbital forcing than the WAIS, lacking a clear obliquity signal in its climate records (Patterson et al., 2026).

NOTE: Again, the below paragraph could go into much more detail, particularly about PLISMIP. I'm interested to know if you feel any particular paragraph in this section is too short, or whether they are all too short (or maybe not.)

The regional records provide coherent evidence of significant mass loss from the WAIS and at least some of the marine sectors of the EAIS. However, ice sheet models in the Pliocene Ice Sheet Model Intercomparison Project (Dolan et al., 2012) were unable to reproduce this mass loss when forced by a Pliocene climate (De Boer et al., 2015; Dolan et al., 2018). Most (but not all) models simulated a collapse of the WAIS, but this was often offset or even exceeded by accumulation over East Antarctica. The theory of marine ice cliff instability (Section 1.3.6) was first popularised by its ability to dramatically increase Antarctic sea level contribution in the mPWP (Pollard et al., 2015). However, it was not required to reach the lower range of Pliocene sea level when accounting for modelling and reconstruction error (Edwards et al., 2019). Other models were later able to simulate higher mass loss from Antarctica without MICI, including from the marine sectors of East Antarctica (Golledge et al., 2017b; Blasco et al., 2024). Furthermore, a grain size proxy that avoids the high uncertainties related to δ^{18}O suggests that sea level oscillated by 4.1–20.7 m over Pliocene glacial-interglacial cycles (Grant et al., 2019; Grant and Naish, 2021). If taken as a deviation from present-day sea level, the upper bound would still require mass loss from marine sectors of East Antarctica, but not from terrestrial sectors.

This progress in both models and palaeoclimate records has made significant progress towards closing the gap between them. Both support a dynamic WAIS that collapsed in climates not so different from that projected by the end of this century. Marine sectors of the EAIS may also be vulnerable to anthropogenic warming, but the palaeoclimate records cannot tell us anything about the pace of retreat — whether it would occur on multi-century timescales. The uncertainties in sea level reconstructions are still very high, but the changes to Antarctica are very large, making them a natural complement to observations in the modern era (with small uncertainties but small changes) when assessing the results of ice sheet models.

1.3 Ice sheet dynamics

1.3.1 Ice rheology*

NOTE: I may want a section here to talk about the polycrystalline structure of ice, as this is relevant to the flow law and the uncertain value of Glen's exponent, but perhaps this is getting too into the weeds.

1.3.2 Ice-atmosphere interactions*

NOTE: Not particularly relevant to my thesis given that we take SMB directly from GCMs. However, a quick review of lapse rates might be useful when discussing why we leave them out later. Anything else in ice-atmosphere that would be good to talk about?

1.3.3 Ice-ocean interactions

871 words

The Southern ocean is a complex system of different water masses that occupy certain depths in the water column, varying significantly in both space and time. We refer the reader to Wang et al. (2025) for a full review, but we highlight several important water masses below that are significant for the Antarctic Ice Sheet.

The uppermost layer of the Southern Ocean, referred to as Antarctic Surface Water (AASW) is governed by turbulent mixing, driven mainly by wind stresses at the surface. The ocean surface is very close to the local freezing point of sea water, approximately -1.8°C. These temperatures circulate through the mixed layer, which varies seasonally in thickness due to changes in the winds. The mixed layer is deepest in winter, reaching depths of 100–200 m or more, but retreats in summer, leaving behind a cold layer known as Winter Water (WW). This layer maintains its cold temperatures whilst the surface waters above it warm in summer. The salinity of AASW is between **TODO: value** psu, less than the global average of **TODO: value** psu. This is for two reasons: firstly, the Southern ocean has a net positive freshwater exchange with the atmosphere, due to receiving more freshwater in precipitation than it loses in evaporation; secondly, it receives freshwater from the melting of sea ice and the Antarctic ice sheet. This low-salinity is inherited by WW, though summer surface water is fresher due to that year's summer ice melt.

The Southern Ocean is also a confluence of deep water masses that are transported through the global overturning circulation (Talley, 2013). Deep water from the Pacific, Indian, and North Atlantic Oceans mix to form Circumpolar Deep Water (CDW). From these water masses, CDW inherits warm temperatures of **TODO: value** °C and high salinity of **TODO: value** psu. It occupies the water column at depths of up to 3000 m, but shoals further South due to regional upwelling. The prevailing Westerly winds that circle the Antarctic continent create a meridional Ekman flow, driving surface waters equatorward and drawing up deep waters to replace them. In these regions, CDW sits just below WW, forming a steep temperature and salinity gradient between the two water masses that appears in ocean depth profiles.

deepest in the winter due to intense winds, reaching depths of 100–200 m or more. This creates the uppermost layer of the Southern Ocean, referred to as Winter Water (WW).

is occupied by very cold surface waters, with temperatures close to the local freezing point of sea water, approximately -1.8°C. This water tends also to be very fresh, due to the freshwater fluxes from sea ice and the Antarctic ice sheet. This maintains a degree of stratification

The Southern Ocean is a confluence of water masses transported through the global overturning circulation. Deep water from the Pacific, Indian, and North Atlantic Oceans mix to form Circumpolar Deep Water (CDW) at depths of up to 3000m (Talley, 2013). CDW is characterised by warm temperatures and high salinity

which circulates through the Antarctic Circumpolar Current (ACC) at depths of up to 3000 m (Talley, 2013). CDW is characterised generally by much warmer temperatures and higher salinity than the surface waters above it (Wang et al., 2025). It can be divided into 'upper' and 'lower' water masses with distinct characteristics. Upper CDW includes contributions from Indian and Pacific deep waters, which are warm, oxygen-poor, and nutrient-rich; Lower CDW contains more North Atlantic Deep Water, giving it characteristically high salinity (Whitworth et al., 2013). Upper CDW shoals as it moves Southwards, eventually mixing with surface waters at the Southern boundary of the ACC. However, lower CDW is dense enough to remain below the mixed layer and extend further towards the Antarctic continent (Orsi et al., 1995). The melting of Antarctic ice shelves is predominantly a question of whether this CDW is able to access the continental shelf that surrounds Antarctica (Wang et al., 2023).

Access to the shelf is limited by the Antarctic Slope Front (ASF), a density front along the shelf break, and a corresponding westward Antarctic Slope Current (ASC) (Jacobs, 1991). The ASF acts as a barrier between CDW and the cooler, fresher shelf waters that occupy ice shelf cavities around Antarctica. The structure of the ASF is highly variable across different sectors of Antarctica and is sensitive to local wind stresses (Spence et al., 2014) and freshwater fluxes (Si et al., 2023). These local conditions give rise to three distinct ice shelf cavity regimes: cold and fresh cavities, cold and saline cavities, and warm and saline cavities.

In Warm and saline cavities such as those in the Amundsen Sea, a weak ASF allows CDW to encroach onto the shelf through

The ASF is weakest in the Amundsen Sea, where the Eastward Antarctic Circumpolar Current dominates over the westward ASC (). This Eastward flow is associated with a polewards Ekman transport in the layer below, which helps to transport CDW onto the continental shelf (Klinck and Dinniman, 2010). This oc

The Southern boundary of the ACC flows very close to the continental shelf in the Amundsen and Bellinghausen Seas, but is deflected away from the Antarctic continent by cyclonic gyres in the Weddell and Ross Seas (Orsi et al., 1995). Temperatures in the gyres are generally lower than those found in ACC (Wang et al., 2023), but some heat is transported to the shelf break along the Eastern boundary of each gyre via entrained CDW (Assmann and Timmermann, 2005; Ryan et al., 2016).

have strengthened and shifted polewards as a response to greenhouse gas emissions and ozone depletion (Waugh et al., 2013; Lee and Feldstein, 2013).

1.3.4 Buttressing, calving and hydrofracture*

NOTE: Placemarker section in case I have time to discuss this in more detail.

1.3.5 Subglacial hydrology*

NOTE: Placemarker section in case I have time to discuss this in more detail.

1.3.6 Ice sheet instability

Since the 1970s, it has been suggested that the West Antarctic ice sheet may undergo rapid deglaciation in a warming climate (Mercer, 1978). This has evolved into the theory of Marine Ice Sheet Instability (MISI), which states that the grounding line position of a marine ice sheet on a retrograde slope (i.e. the bed becomes deeper towards the centre) is inherently unstable (Schoof, 2007). The fundamental principle of MISI is that discharge at a glacier's terminus is proportional to the glacier thickness at its grounding line. The retreat of the grounding line down a retrograde slope increases thickness, which in turn increases discharge, which causes the grounding line to retreat further, creating a feedback loop. In practice, this effect is mitigated by ice shelves, which buttress the ice sheet, generating sufficient friction to stabilise the grounding line (Joughin and Alley, 2011). However, Antarctica's ice shelves are also thinning as a result of climate change. In 2002 the Larsen A and B ice shelves on the Antarctic peninsula collapsed over just a few days (Shepherd et al., 2003).

The West Antarctic Ice Sheet is a strong candidate for undergoing MISI, since it overlies a deep bathymetric basin up to 2000m below sea level, giving it an overall retrograde slope. This is supported by satellite measurements of Antarctic mass balance, which identify the Amundsen Sea Embayment in West Antarctica as the primary region for ice loss in recent decades (Smith et al., 2020; Velicogna et al., 2020; Otosaka et al., 2023). Pine Island glacier, which terminates in the Amundsen Sea, shows evidence of Circumpolar Deep Water at its grounding line, retreat down a retrograde slope, thinning of its ice shelves, and an acceleration in its discharge (Joughin et al., 2003; Jenkins et al., 2010; Jacobs et al., 2011; Favier et al., 2014). It has been proposed that West Antarctica could be approaching a 'tipping point', in which MISI could push the ice sheet into a dramatically new state with only a small change in forcing (Pattyn and Morlighem, 2020; Rosier et al., 2021; Armstrong McKay et al., 2022). Regional ocean models that are able to resolve the temperature in ice shelf cavities find that further warming is unavoidable under any future scenario (Naughten et al., 2023), indicating that collapse of the West Antarctic ice sheet may now be inevitable.

Compared to the WAIS, the EAIS has been broadly considered to be relatively stable. However, the EAIS, whilst mostly above sea level, does include some marine sectors which may be vulnerable to MISI (Armstrong McKay et al., 2022). As with the Amundsen Sea Embayment, warm Circumpolar Deep Water has been observed infiltrating the ice shelf cavities and driving thinning of the Totten glacier, which drains part of the Aurora subglacial basin (Greenbaum et al., 2015; Aitken et al., 2016; Rintoul et al., 2016; Silvano et al., 2017). The ice shelf at the terminus of the Cook glacier, which drains a large proportion of the Wilkes basin, collapsed between 1973–1989, and basal thawing has recently been observed along the nearby Adélie-George V coast (Miles et al., 2018; Dawson et al., 2024). However, warming over the Southern ocean also drives increased precipitation over East Antarctica (Nicola et al., 2023), which has the potential to offset some or all of the mass loss in the marine sectors. Uncertainty in the overall mass balance is large, but a synthesis of multiple modelling and observation studies suggests that it will remain approximately in balance this century, but has the potential to cause multiple metres of sea level rise on multi-century timescales (Stokes et al., 2022).

Other than marine ice sheet instability, additional mechanisms have been proposed that could drive rapid deglaciation in marine segments of Antarctica. Marine Ice Cliff Instability (MICI;Bassis and Walker 2012) is the theory that marine-terminating ice cliffs could be liable to rapid retreat under certain conditions. Models that include MICI yield significantly greater sea level rise than those without (DeConto and Pollard, 2016; DeConto et al., 2021). However, the inclusion of visco-elastic effects in such models indicate that ice cliffs could be more stable than previously thought (Clerc et al., 2019; Bassis et al., 2021). Bathymetric grooves in the ocean floor could hint at evidence of MICI (Wise et al., 2017), but it has yet to be observed in the modern era, nor is it necessary to achieve the sea level change seen in past climates (Edwards et al., 2019). Similarly, corrugation ridges along the ocean floor suggest a 'buoyancy-driven' mechanism for ice sheet retreat, up to hundreds of metres per day along low-gradient slopes (Batchelor et al., 2023). However, like MICI, it has yet to have been observed in the modern climate.

1.3.7 Solid Earth effects

On short to medium timescales, ice-ocean interactions and ice sheet stability are the main dynamics to consider regarding future sea level rise. However, on longer timescales spanning multiple centuries or millennia, additional effects relating to the interaction between the ice sheets, the solid Earth, and its gravitational field, become relevant. These effects are collectively referred to as Glacial Isostatic Adjustment (GIA;Whitehouse 2018, and references therein). As an ice sheet retreats, the load that it exerts on the Earth's crust decreases, inducing a visco-elastic response which uplifts the land surface due to upwards pressure in the asthenosphere. GIA is currently observed in regions that experienced deglaciation following the last ice age, recording uplift of over 10 mm a^{-1} in some parts of North America and Europe (Sella et al., 2007; Lidberg et al., 2010). As evidenced by these regions, which have been without an ice sheet for over 8,000 years, the response to glacial unloading is not instantaneous and can be drawn out for millennia.

GIA has the capacity to significantly influence shoreline migration and bedrock topography over long timescales, which in turn feeds back into the dynamics driving ice sheet growth or retreat. For example, the uplift of bedrock following unloading could limit the accessibility of Circumpolar Deep Water or reverse/intensify the retrograde slope in a region susceptible to Marine Ice Sheet Instability. Larour et al. (2019) find that GIA mitigates 27% of projected sea level rise over 500 years using a model of the Thwaites glacier in which it is included. Similarly, Kachuck et al. (2020) show that GIA could reduce ice mass loss from the Pine Island Glacier by 30% over 150 years. These modelling studies show that GIA can have an impact on even sub-millennial timescales.

1.4 Ice sheet modelling

68 words

Numerical ice sheet models are important tools for Antarctic science, both for understanding the mechanisms behind recent ice sheet change (e.g. Payne et al. 2004; Gudmundsson et al. 2019; Bradley et al. 2025), exploring possible ice sheet configurations in the deep past (e.g. De Boer et al. 2015) and for making projections of future ice sheet change (Seroussi et al. 2020, 2024, among many others). This section reveiws the construction of a general ice sheet model and the important choices involved in simulating the Antarctic ice sheet.

1.4.1 Stokes flow

200 words

Practical ice flow modelling is done using a hierarchy of approximations to incompressible Stokes flow, which is itself the low Reynolds number limit of the Navier-Stokes equations. In this limit, inertial terms are ignored, and flow is governed by a balance between pressure gradients, viscous stresses, and gravitational driving stresses. These are solved alongside a constitutive relation, which defines the ice rheology (see Section 1.4.2). It is also necessary to impose boundary conditions at the base and surface of the ice sheet. For a rigid boundary (the interface at the bed), this might take the form:

$$\mathbf{u}_b = F(\boldsymbol{\tau}_b), \quad \mathbf{u}.\mathbf{n} = -m_b, \tag{1.1}$$

where $\mathbf{u}$ is the ice velocity field, $\mathbf{u}_b$ is the velocity component parallel to the bed, $\mathbf{n}$ is the normal to the bed, $\boldsymbol{\tau}_b$ is the basal shear stress, F is a sliding law (see Section 1.4.4), and m_b is the basal melt rate (in velocity units). For a free surface (the interface with the atmosphere or ocean), dynamic and kinematic boundary conditions are applied to specify the boundary stresses and surface evolution, respectively. Some model setups include an advection-diffusion equation for temperature, which couples to the momentum (Stokes) equations through the constitutive relation (Eq. 1.2). However, assuming isothermal ice (i.e. fixed temperature) is often a reasonable approximation over short timescales.

1.4.2 Constitutive relation

227 words

The constitutive relation describes the relationship between the stress and deformation of a material. For ice, it is common to use Glen's flow law (Glen, 1955):

$$\dot{\epsilon}_{ij} = A\tau^{n-1}\tau_{ij}, \quad \tau = \sqrt{\frac{1}{2}\tau_{ij}\tau_{ij}}, \tag{1.2}$$

where ϵ_{ij} is the strain rate tensor, τ_{ij} is the deviatoric stress tensor, $A(T)$ is a temperature-dependant rate factor, and n is a scalar exponent ("Glen's exponent"). Early work to estimate the value of n for polycrystalline ice suggested that $n \approx 3$ (Weertman 1983 and references therein). Subsequently, $n = 3$ became standard across a wide range of ice sheet models (e.g. Larour et al. 2012; Cornford et al. 2013; Lipscomb et al. 2019). However, Goldsby and Kohlstedt (2001) demonstrated that creep can be divided into multiple regimes, occurring through different mechanisms within the polycrystalline structure. The appropriate value of n depends on the dominant creep mechanism, which in turn depends on the stress. **NOTE: I could talk here about typical driving stresses in Antarctica and the corresponding value $n = 1.8$, but my attempts at this have gotten a bit bogged down and wordy, so have removed for better flow**. Other studies suggest that observations of Antarctic ice streams and ice shelves may be better explained by a higher value of n (e.g. Cuffey and Kavanaugh 2011; Millstein et al. 2022). Ice sheet models that have explored this found Antarctic sea level contribution to be sensitive to the choice of n, with higher values leading to greater sea level rise (Rosier et al., 2025; Getraer and Morlighem, 2025; Martin et al., 2026). **NOTE: Is signposting like "We will explore this uncertainty in chapter 2" (or similar) appropriate in these** situations?

TODO: Cuffey and Paterson and have a good section on this — worth a brief rewriting after reading

1.4.3 Approximations in ice flow models

39 words

Full Stokes modelling is computationally expensive and so is rarely deployed for continent-scale ice sheet modelling (with some notable exceptions, e.g. Seddik et al. 2012). Approximations are typically applied to match the requirements of a particular ice-dynamical regime.

1.4.4 Basal sliding

1.4.5 Surface mass balance*

1.4.6 Sub-shelf melting

1.4.7 Glacial isostatic adjustment

1.5 Projections of the Antarctic ice sheet

1.5.1 Climate context

684 words

Climate projections in AR6 were informed by the Coupled Model Intercomparison Project Phase 6 (CMIP6; Eyring et al. 2016), a coordinated set of climate experiments organised by the World Climate Research Programme. AR6 structured its climate projections as a collection of storylines, each describing a narrative of how society develops over the 21st century (O'Neill et al., 2016).

In the previous assessment report (CMIP5; Taylor et al. 2012), projections followed the Representative Concentration Pathways (RCPs; van Vuuren et al. 2011), which were defined only by their greenhouse gas emissions. AR6 developed this by introducing the Shared Socioeconomic Pathways (SSPs; Riahi et al. 2017), which considered the technological development, international collaboration, and socioeconomic conditions that shape our ability to mitigate and adapt to climate change. For example, SSP1 is a sustainable pathway with a focus on human wellbeing, whereas SSP5 is a growth-oriented pathway with high fossil fuel consumption. Integrated Assessment Models then combined these with specific climate policy targets to produce their emissions trajectories. These targets were chosen to roughly align the radiative forcing of the SSPs with the RCPs: SSP1-2.6 with the low-emissions RCP2.6 and SSP5-8.5 with the high-emissions RCP8.5. Between them, they represent the spread in possible emission futures, though SSP5-8.5 is now considered unlikely given the pace of global decarbonisation (Hausfather, 2025).

AR6 projected a late 21st century global climate that is 0.6–2.0 °C warmer than the 1995–2014 climatology in SSP1-2.6 and 2.7–5.7 °C warmer in SSP5-8.5 (Lee et al., 2021). The change to the Antarctic climate is similar but more uncertain, between 0.1–2.9 °C warmer in SSP1-2.6 and 1.7–5.6 °C in SSP5-8.5. It projected continued warming and freshening of Southern Ocean surface and intermediate waters, which increases evaporation at the ocean surface and is linked to higher precipitation over Antarctica (Fox-Kemper et al., 2021; Douville et al., 2021). The models predict an average increase in precipitation of 5.5% per degree of warming, exceeding 10% over the East Antarctic Plateau (Nicola et al., 2023). This increases the total surface mass balance of the Antarctic ice sheet, but is partially offset under high warming by enhanced surface melting of Antarctic ice shelves (Kittel et al., 2021).

AR6 had *low confidence* in projected changes in the supply of Circumpolar Deep Water to Antarctic ice shelves due to numerous limitations of the CMIP6 models: firstly, they did not resolve ice shelf cavities, meaning they could not capture the cavity circulation, mixing, or the bathymetric features that are important for modelling basal melt rates (Burgard et al., 2022); secondly, their resolution was too coarse to resolve the eddy processes that help to transport CDW onto the continental shelf (Wang et al., 2023); thirdly, persistent warm biases in the Southern Ocean across multiple generations of climate models raise concerns about the validity of projected changes (Barthel et al., 2020); and lastly, climate models at the time included only a static ice sheet, ignoring the ice-ocean meltwater feedbacks that can affect the strength of CDW reaching the base of ice shelves (section 1.3.3). Instead, high-resolution ocean models, which resolve ice shelf cavities, eddies, and include meltwater fluxes, are required to confidently project changes in ice shelf basal melt (Yung et al., 2026). Such simulations of the Amundsen Sea project an enhanced transport of CDW onto the continental shelf and accelerated melting of its ice shelves (Jourdain et al., 2022; Naughten et al., 2023). Naughten et al. (2023) show that this is nearly independent of the future emissions scenario, implying that these 21st century changes are now unavoidable. The crossing of a tipping point — in which CDW flushes a presently 'cold' ice shelf cavity and flips it to a 'warm' state — is also possible this century for ice shelves such as the Filchner-Ronne (Hellmer et al., 2012, 2017; Song et al., 2025) and the Ross (Siahaan et al., 2022), though the timing and likelihood of this transition remains uncertain and model-dependent (Naughten et al., 2021; Comeau et al., 2022; Kusahara et al., 2023)

The climate of the late 23rd century is strongly dependent on future emissions, with global temperatures 6.6–14.1 °C higher than the preindustrial era in SSP5-8.5, but only 0.6–1.4 °C higher in SSP1-2.6 (Lee et al., 2021). Under high emissions, the Antarctic climate would be radically different than today, with surface melt possibly overtaking snow accumulation to drive an overall negative surface mass balance (Coulon et al., 2024). The Southern Ocean could become virtually free of sea ice, with CDW occupying most ice shelf cavities and basal melt rates increasing more than 10-fold (Mathiot and Jourdain, 2023). It has been suggested that Antarctica could undergo "Greenlandification" (Mottram et al., 2025), mimicking the conditions that have accelerated Greenland Ice Sheet mass loss in the present day.

1.5.2 21st century projections

984 words

The Ice Sheet Model Intercomparison Project for CMIP6 (ISMIP6; Nowicki et al. 2016) was a CMIP6-endorsed experiment that specifically explored changes in the ice sheets and their contribution to sea level rise. ISMIP6 brought together multiple ice sheet modelling communities to conduct experiments under a consistent protocol (Nowicki et al., 2020). This included an experiment exploring the sensitivity of idealised experiments to the ice sheet initial state (InitMIP; Seroussi et al. 2019) and experiments exploring the Greenland and Antarctic Ice Sheet response to a range of 21st century emission scenarios (Payne et al., 2021). At the time of ISMIP6, the climate model outputs from CMIP6 were not yet available, so ice sheet models instead used CMIP5 forcing for a range of RCPs. Modelling groups were asked to run a control simulation under a fixed baseline climate, which would be subtracted from the main sea level projections. They could also optionally include the collapse of Antarctic ice shelves, prescribed based on the rate of surface melting, although 21st century collapse was mostly limited to the Antarctic Peninsula and Totten Glacier

(Seroussi et al., 2020).

The results of ISMIP6 yielded a Antarctic sea level contribution of between -1.4–15.5 cm in RCP2.6 and -7.8–30 cm in RCP8.5, with ice shelf collapse increasing the maximum sea level contribution by 2.8 cm (Seroussi et al., 2020). Most of this mass loss originates from the West Antarctic Ice Sheet, with limited mass gain over East Antarctica (Seroussi et al., 2020). However, the highest values were obtained using the most sensitive climate model and basal melt calibration, which were outliers relative to the rest of the ensemble (Edwards et al., 2021). In addition, the control experiment yielded sea level contribution between -13–14.2 cm (Seroussi et al., 2020). This spread is similar to that of the RCPs, underlining the strong influence of non-forcing factors such as initial state and model structural uncertainty, also seen in InitMIP (Seroussi et al., 2019). Edwards et al. (2021) fit a Gaussian process emulator to the ISMIP6 ensemble, using changes in global mean temperature (among other inputs) as training data. This allowed them to combine the ISMIP6 ensemble with the SSPs (via the FaIR simple climate model) to produce probability distributions of future sea level. Emulation has the added benefit of considering information from the whole ensemble and not just the extremes, which considerably narrows the high-end projections (Edwards et al., 2021). This removes the scenario-dependence of the projections, with 5–95% confidence intervals of -1–19 cm in SSP1-2.6 and 0–18 cm in SSP5-8.5 (Fox-Kemper et al., 2021). **NOTE: Why do the AR6 numbers here not match the values from the original edwards2021 paper?**

The Linear Antarctic Response Model Intercomparison Project (LARMIP-2), took a different approach to 21st century sea level projections, using linear response theory to estimate the Antarctic sea level response to ocean warming. Unlike the process-based modelling of ISMIP, they conducted a series of idealised experiments which could then be statistically mapped onto the RCPs (Levermann et al., 2020) or SSPs (Fox-Kemper et al., 2021) through changes in global mean temperature. This simpler protocol resulted in a higher number of participating ice sheet models, but it also ignored surface mass balance changes or non-linear effects such as MISI. Even adjusting for surface mass balance, LARMIP-2 obtained a higher 5–95% range than the emulated ISMIP6 ensemble: 3–41 cm in SSP1-2.6 and 1–57 cm in SSP5-8.5 (Fox-Kemper et al., 2021). This was largely due to the high sensitivity of ice shelf melt in LARMIP-2, which corresponds well with melt rates in the Amundsen Sea, but overestimates melt in most other ice shelf cavities (Reese et al., 2020). Nonetheless, AR6 did not make any judgement on whether the ISMIP6 or LARMIP-2 projections were more realistic, conservatively combining LARMIP-2 with the emulated ISMIP6 to yield a 5–95% confidence interval of -1–41 cm in SSP1-2.6 and 0–57 cm in SSP5-8.5 (Fox-Kemper et al., 2021).

A common theme among projections of the Antarctic ice sheet is the competition between the ocean and atmospheric drivers of mass change, the balance of which can determine the sign of Antarctica's sea level contribution (Seroussi et al., 2020; Edwards et al., 2021). These competing effects are manifest in the different responses of the West and East Antarctic Ice Sheets, reflecting the different processes that govern these regions. The marine WAIS is highly influenced by its ice shelves and its response is therefore dominated by ocean warming, whereas the EAIS has a mixed responses of mass loss from its marine sectors and mass gain over the East Antarctic Plateau (Seroussi et al., 2020, 2023; O'Neill et al., 2025). Uncertainty in the ensemble is overall dominated by the choice of ice sheet model, but drivers of uncertainty can vary even on a glacier-scale (Seroussi et al., 2023).

Since AR6, several studies have given more context to the results of ISMIP6 and LARMIP-2. For example, the linear basal melt parameterisation used in LARMIP-2 has been shown to perform poorly relative to the quadratic parameterisation used in ISMIP6 or other basal melt parameterisations (Burgard et al., 2022). On the other hand, the basal melt sensitivity used for high-end projections in ISMIP6 compares poorly with coupled ocean-ice shelf models, suggesting that these were overestimated (Jourdain et al., 2022). Improvements have been made to ice sheet or climate models, including several climate models that have implemented a coupled dynamic ice sheet (Muntjewerf et al., 2021; Siahaan et al., 2022; Park et al., 2023; Goelzer et al., 2025; Sadai et al., 2025). Only a few of these have made 21st century projections with a coupled Antarctic Ice Sheet, but they highlight the complex meltwater feedbacks that can both enhance and inhibit mass loss (Park et al., 2023). Overall, these feedbacks tend to reduce Antarctica's mass loss (Park et al., 2023; Sadai et al., 2025) or even yield a mass gain (Siahaan et al., 2022), but the limited number of these studies means that this balance is still uncertain and it may change on longer timescales.

An important conclusion of AR6 was that the Antarctic Ice Sheet is *likely* to continue losing mass throughout this century, irrespective of future emissions scenario (Fox-Kemper et al., 2021). It emphasised that whilst the scenarios with higher warming drive a different response in the ice sheet, there was no consensus on which of the competing effects driving mass loss or mass gain would emerge as dominant over the 21st century. This makes the influence of 21st century emissions over Antarctic mass loss difficult to assess, demonstrating the need for longer-timescale projections if we are to fully understand the consequences of present-day climate policy on future sea level rise (Lowry et al., 2021).

1.5.3 Multi-century projections

783 words

As explained in the introduction to this chapter, ISMIP6 did not make projections of sea level contribution beyond 2100, with multi-century projections relying on extrapolation, expert elicitation, and disparate modelling studies to make the AR6 assessment. This is largely due to a lack of appropriate forcing, since state of the art climate model simulations did not generally extend past 2100 at this time. The RCPs were extended to 2500 as described in Meinshausen et al. (2011). However, multi-century projections in CMIP5 were reliant on models of intermediate complexity (Zickfeld et al. 2013). These were less sophisticated than the more complex General Circulation Models (GCMs) and do not output the necessary climate variables to run an ice sheet model. Multi-century ice sheet projections then either fixed the climate after 2100 (Bindschadler et al., 2013; Lowry et al., 2021; Lipscomb et al., 2021; Chambers et al., 2022), used an idealised forcing (Martin et al., 2019; Garbe et al., 2020), or constructed an appropriate forcing through statistical methods (Golledge et al., 2015; Bulthuis et al., 2019; DeConto et al., 2021; Greve et al., 2023). For example, Golledge et al. (2015) and Bulthuis et al. (2019) extend the 2100 CMIP5 GCM projections by scaling them linearly with temperature, which is derived from the CMIP5 intermediate complexity models. This provides a proxy for RCP-based multi-century climate simulations, but assumes that the relationship between temperature and other climate variables stays constant with time. Nonetheless, these studies highlight the importance of multi-century forcings, which can double sea level projections relative to a forcing that's fixed after 2100 (Seroussi et al., 2024; Greve et al., 2023).

Since ISMIP6, several climate models (without coupled ice sheets) have now ran climate simulations up to 2300 using extended SSPs (Meinshausen et al., 2020). This has, for the first time, allowed multi-century ice sheet projections without the need for statistical extrapolation of climate forcing. Seroussi et al. (2024) conducted an ISMIP6-style experiment designed for multi-century projections of the Antarctic Ice Sheet. The experiment focused mainly on a high emissions future scenario, with only 2 out of 14 recommended forcings representing a low emissions pathway. The results, gathered from 16 participating ice flow models, highlight the acceleration of mass loss from Antarctica that occurs after the end of the 21st century, up to 4.4 m sea level equivalent by 2300. Ice shelf collapse affects most regions with floating ice by the end of the 23rd century, increasing sea level contribution by 1.1 m on average. The experiment produced dramatic changes in the WAIS by 2300, with grounding line retreat along the Siple and Weddell coasts likely to exceed 200 km. Nearly half of models initiate a retreat of Thwaites Glacier, finding stability near to a pinning point 400 km behind its current grounding line. The EAIS changes from a net mass gain over the 21st century to a net mass loss by the end of the 23rd century, again originating from its marine sectors. However, this is muted in comparison to studies that include MICI, which leads to significant ungrounding in the Wilkes and Aurora basins (DeConto et al., 2021; Gomez et al., 2024). The response under low emissions is less well explored — the limited simulations by Seroussi et al. (2024) project a sea level contribution between -0.37 and 0.62 m, but other studies suggest that Thwaites Glacier could continue to retreat, raising sea level by up to 2 m (Klose et al., 2024; Coulon et al., 2024, 2025).

Uncertainty in multi-century projections remains very high, but several efforts have been made to characterise and quantify it (Bulthuis et al., 2019; Coulon et al., 2024, 2025). As with 21st century projections, the ocean is the primary driver of overall mass loss throughout the 22nd and 23rd centuries due to its strong control on the WAIS (Bulthuis et al., 2019; Greve et al., 2023; Coulon et al., 2024). However, the East Antarctic ice sheet response is increasingly driven by atmospheric changes after 2100 and can be highly dependent on the forcing climate model (Coulon et al., 2025). Significantly stronger forcing over three centuries can overcome some of the ice sheet model structural uncertainty that dominated 21st century projections (Coulon et al., 2025), but there is still disagreement when considering the full range of ISMIP models (Seroussi et al., 2024). A stronger scenario-dependence also emerges by the end of the 23rd century, with a clean separation of the uncertainties between high and low emission projections (Fox-Kemper et al., 2021; Greve et al., 2023; Coulon et al., 2024; Klose et al., 2024), although wider sampling of uncertainty can still result in a limited overlap (Coulon et al., 2025).

Past the 23rd century, projections are again limited by the available forcing, but long-term simulations highlight the continued mass loss from Antarctica that occurs well after 2300 (Golledge et al., 2015; Bulthuis et al., 2019; Klose et al., 2024; Coulon et al., 2024, 2025). Committed sea level from Antarctica is very sensitive to the forcing climate model, but can reach up to 40 m (Klose et al., 2024). In SSP5-8.5, the WAIS, as well as a significant fraction of the Wilkes basin, can become ungrounded by year 3000 (Bulthuis et al., 2019; Coulon et al., 2024, 2025), possibly followed by all marine sectors of East Antarctica on multi-millennial timescales (Klose et al., 2024). High mitigation can prevent substantial mass loss, but cannot rule out collapse of the Amundsen Sea Sector and committed sea level rise of up to 6 m (Klose et al., 2024). These millennial-scale projections highlight the long response time of ice sheets to high forcing — particularly East Antarctica, which is barely ungrounded by 2300 in models without MICI, but which can experience widespread collapse with no further change in forcing.

Chapter 2

Methods

This is the methods chapter.

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